CROSSSAFE · FEASIBILITY ASSESSMENT

CrossSafe feasibility assessment

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	CRS-FEA-001	Version	0.1
Status	Draft	Date	2026-10-31
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/crosssafe		
Source	docs/08-feasibility.md @ d9f28be		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First feasibility assessment (kit 1.9.1 upgrade) from the TRL 3 calculations and the model

CrossSafe feasibility assessment

Option A is feasible on paper as a bought-parts build: the model of 117 parts passes the D3 check, 13 of 19 requirements are met on paper or by design, and what remains (visibility, detection, install time, flash area, cost) needs a trial or an Amish decision. Method: STANDARDS section 23.

Table 1. Feasibility by aspect (option A).

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	The geometry, energy, heat, wind and radio all close on paper; three STEM rows were fixed (S3 charge limit, S5 bar finish, S22 for option C)	High	Cost is $14.00 over its target; flash area At risk	CRS-STM-001, CRS-CAL-001 v0.7
Materials	Steel post, aluminium bar and plates, polycarbonate enclosure, stainless fasteners, EPDM seals; no exotic materials	High	Corrosion in coastal sites; UV on the polycarbonate; to assess	CRS-CAL-001, BOM
Manufacturing and assembly	65 made or bought parts and 47 fasteners; sheet bending, drilling and a few welds; 18 assembly steps	Medium	Welding skill for 5 welds; 120 min install is at the limit (PR-010)	CRS-BLD-001, cad/fidelity.json
Power and controls	20 W panel and 154 Wh battery give 7.7 days; charge to 55 C needed in the hot box	Medium	Battery life at 55 C; radio band differs in the US (915 MHz)	CRS-CAL-001 B, C4
Supply of parts	Signs, LED heads, enclosure, panel and battery are catalogue items	Medium	Prices are indicative; LED heads with visors and a 55 C battery need quotes	BOM
Regulatory and approval	Form follows IA-21; acceptance is the road authority's decision	Low	Whether IA-21 is current, the MUTCD edition and state rules are not checked	CRS-REG-001
Use and acceptance	Passive detection and the button give the pedestrian a trigger; no evidence of effect for this device	Medium	PR-002 and PR-005 not verifiable at TRL 3	CRS-CAL-001

Ratings: High, Medium, Low. Options B and C: B is feasible but costs about $400 and relies on road works (Low-Medium); C is the most feasible to build (40 parts, about $70) and the least proven in effect (CRS-OPT-001).